

[illegible]

It is given that $\cos \theta = \frac{5}{6}$.

- (b) Find, in terms of a and g , an expression for the vertical component of the velocity of P just before it hits the horizontal plane to which the sphere is fixed. [3]

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- (c) Find an expression for the time taken by P to fall from Q to the plane. Give your answer in the form $k\sqrt{\frac{a}{g}}$, stating the value of k correct to 3 significant figures. [2]

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